

INCLUSIVE EMPLOYMENT PLATFORM

Accessible, AI-Assisted Recruitment for Persons with Physical Disabilities

Final Engineering Presentation

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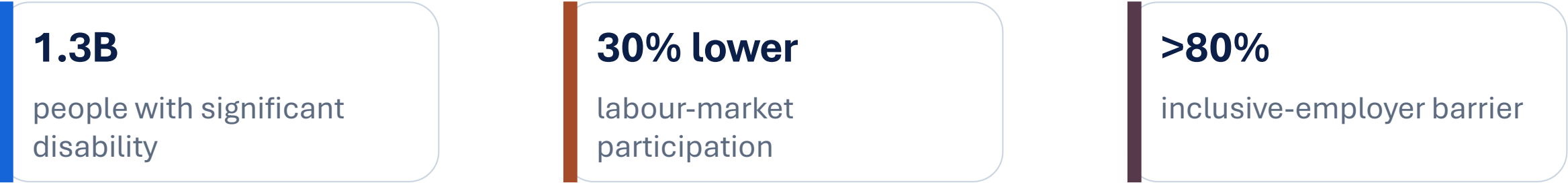
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CONTEXT AND PROBLEM

Capability and opportunity remain disconnected

Inaccessible processes hide ability and reduce trust in task-level decisions.



The engineering problem is not a lack of ability.

It is the distance between accessible recruitment, verified candidate profiles, real job tasks, workplace support and employers prepared to evaluate ability.

Sources retained from the report: WHO; ILO Global Disability Summit 2025; Business Disability Forum / Evenbreak survey.

Related approaches and the remaining gap

The project combines strengths that usually appear in separate solutions.

General portals

- Large reach
- Familiar workflow
- Generic candidate profiles

Inclusive portals

- Targeted visibility
- Employer awareness
- Limited task-level evidence

AI screening

- Speed and scale
- Bias and privacy risks
- Often opaque decisions

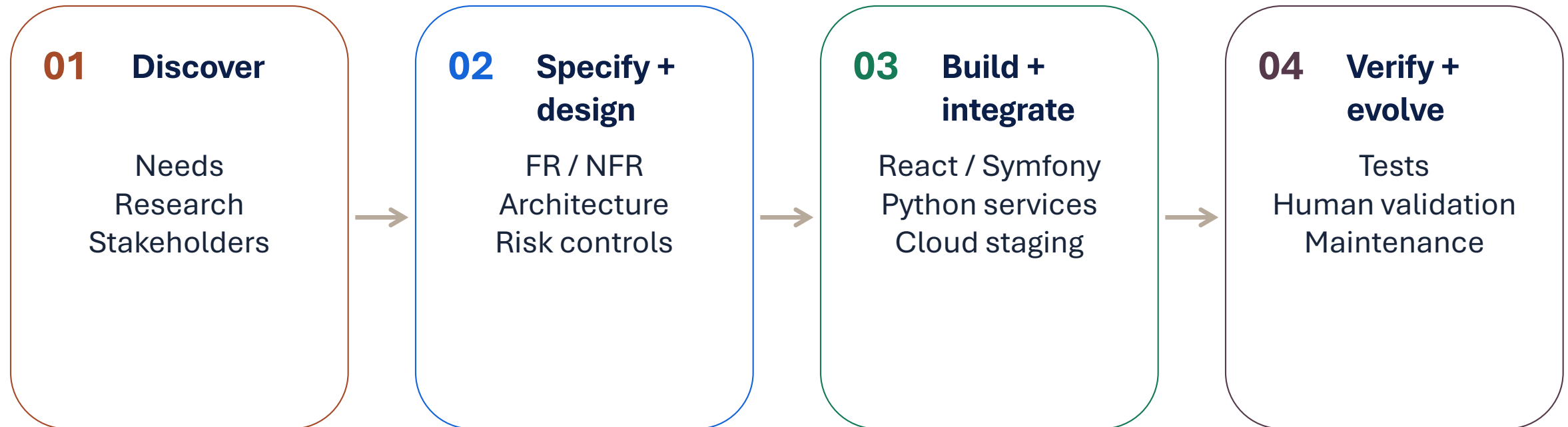
Human coaching

- Contextual judgement
- Adaptation expertise
- Difficult to scale digitally

Join gap: one accessible workflow joining controlled hospitality tasks, candidate abilities, workplace assistance, explainable matching and human review.

An iterative lifecycle with V-model traceability

Evolutionary delivery uses V-model traceability after every material change.



Traceability: stakeholder need -> requirement -> design -> code -> test oracle -> retained evidence

Needs came from people, evidence and prototypes

Stakeholder needs shaped a focused, measurable Phase II delivery.

People

Candidates with varied access needs
Employers and hospitality teams
Verifier and administrator roles

Evidence

Phase I artefacts
WCAG and domain research
Arc en Ciel and JAWS sessions

Techniques

Background study
Scenario observation
Iterative prototype review

4 roles

authorized workspaces

15 roles

hospitality positions

923

catalogue tasks

3

interface languages

Functional requirements implemented by role

Four authorized workspaces coordinate one auditable recruitment workflow.

Candidate

- Verify identity
- Build accessible profile
- Match, explore and apply

Employer

- Publish job or training
- Set tasks and support
- Review applications

Verifier

- Review private evidence
- Approve or reject
- Record an auditable decision

Administrator

- Manage users and catalogue
- Import validated datasets
- Maintain controlled definitions

Nominal, rejected-role and recovery flows were specified and tested.

Quality constraints, standards and measurable evidence

Non-functional requirements were connected to controls and test evidence.

Accessibility

WCAG 2.1 AA target
Keyboard + screen reader
Responsive reflow

Security + privacy

JWT and server RBAC
Private evidence
Consent and erasure

Reliability

Health endpoints
Repeatable p50 / p95
Transparent limitations

Maintainability

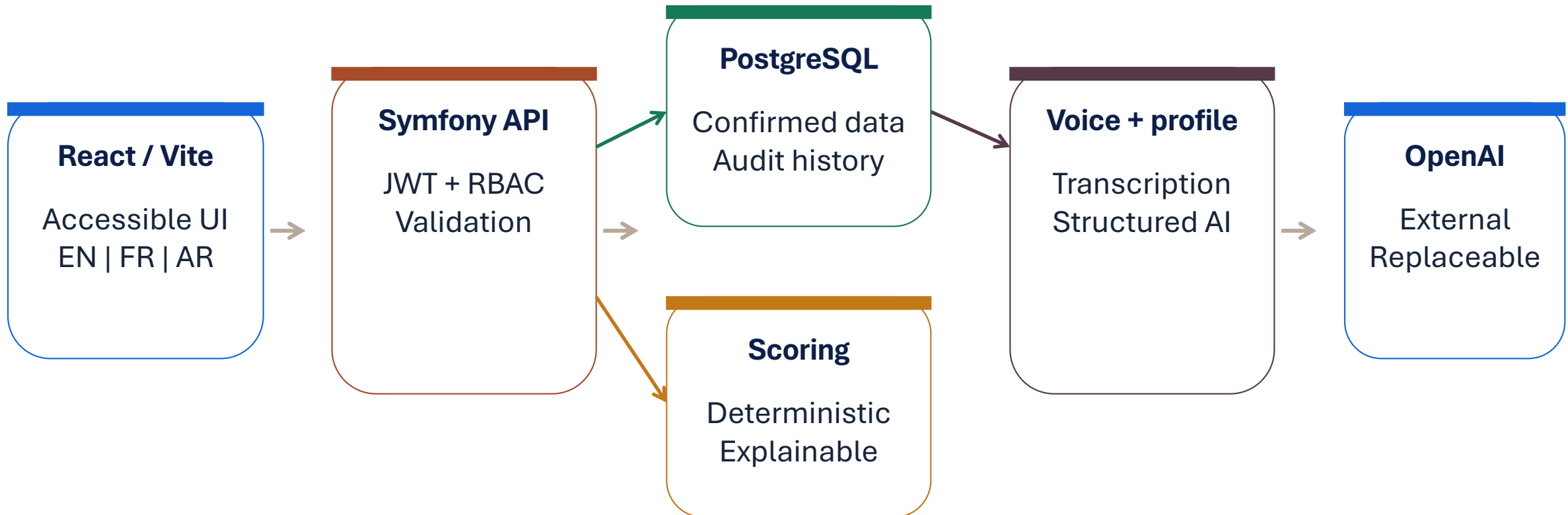
Component boundaries
Versioned policy
Three stable locales

Constraint trade-off: stronger controls and external services can affect latency; correctness and performance are reported separately.

CONCEPTION

Cloud architecture with explicit trust boundaries

React presents; Symfony authorizes; specialist services stay isolated.



Trust boundary: only Symfony may persist confirmed profile data or submit an official scoring payload.

Design decisions control complexity and sensitive data

Patterns address coupling, opacity, unsafe action and lost decision history.

Layered services

Controller - service - repository
Low coupling and cohesive change

Adapters + policy

Replaceable OpenAI boundary
Versioned deterministic scoring

Command registry

Allowlisted voice actions
Role and page authorization

Privacy + audit

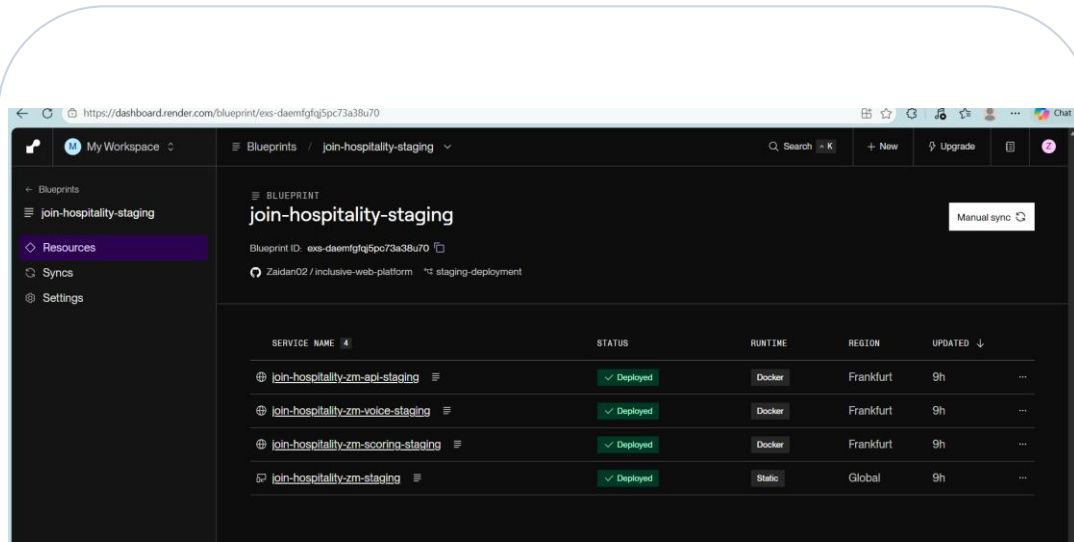
Private document storage
Consent, retention and snapshots

Evidence: 51/51 route probes passed; one oversized-upload defect remains recorded.

PROGRAMMATION ET INTEGRATION

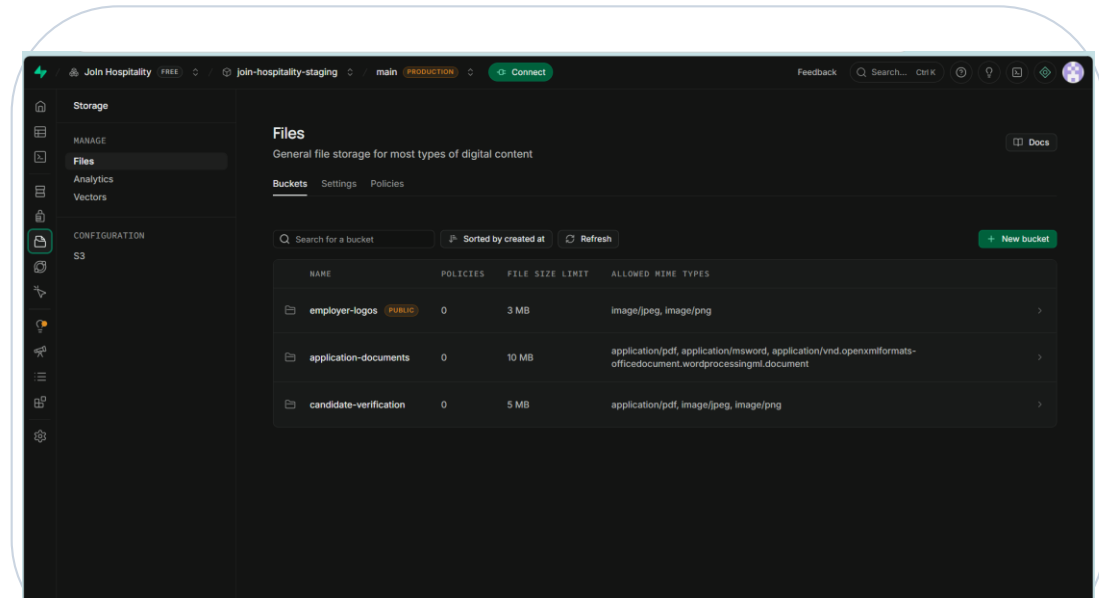
Implemented services running on shared cloud staging

Versioned components deploy independently through infrastructure as code.



Render Blueprint

React static site | Symfony API | scoring |
voice/profile



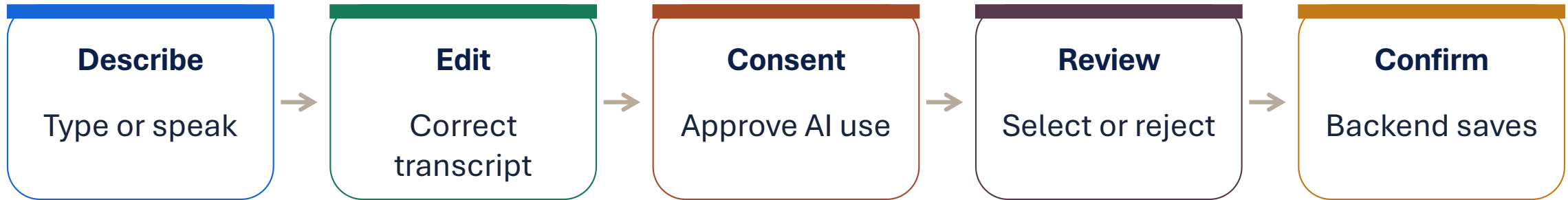
Supabase data layer

PostgreSQL | private evidence | public
employer logos

PROGRAMMATION

AI-assisted profile: multilingual, advisory and reversible

The candidate remains in control from transcription to persistence.



OPTIONAL AI ASSISTANCE Human reviewed

Describe yourself in your own words
Speak or type in English, French, or Arabic. You edit the transcript and approve every suggestion before it is saved.

AI profile input language
English

This may be different from the interface language.

Use microphone Audio is transcribed, then discarded by this application.

Your editable description or transcript

My first name is Maya and my last name is Haddad. My phone number is 70123456 and I live in Beirut. Set my About you text to: I am reliable, patient, and enjoy helping guests. I completed high

ASSISTANCE IA FACULTATIVE Validation humaine

Décrivez-vous avec vos propres mots
Parlez ou écrivez en anglais, en français ou en arabe. Vous pouvez modifier la transcription et approuver chaque suggestion avant son enregistrement.

Langue de saisie du profil IA
Français

Elle peut être différente de la langue de l'interface.

Utiliser le microphone L'audio est transcrit, puis supprimé par cette application.

Votre description ou transcription modifiable

Mon prénom est Maya et mon nom est Haddad. Mon numéro est 70123456 et j'habite à Beyrouth.

مساعدته اختيارية بالذكاء الاصطناعي خاضع لمراجعة بشرية

عرّف عن نفسك بكلماتك الخاصة
تحدث أو اكتب بالإنجليزية أو الفرنسية أو العربية. يمكنك تعديل النص والموافقة على كل اقتراح قبل حفظه.

لغة إدخال ملف الذكاء الاصطناعي
العربية

يمكن أن تختلف عن لغة الواجهة.

استخدام الميكروفون يُحوّل الصوت إلى نص، ثم تخلص منه هذه الصفحة.

وصفك أو النص القابل للتعديل

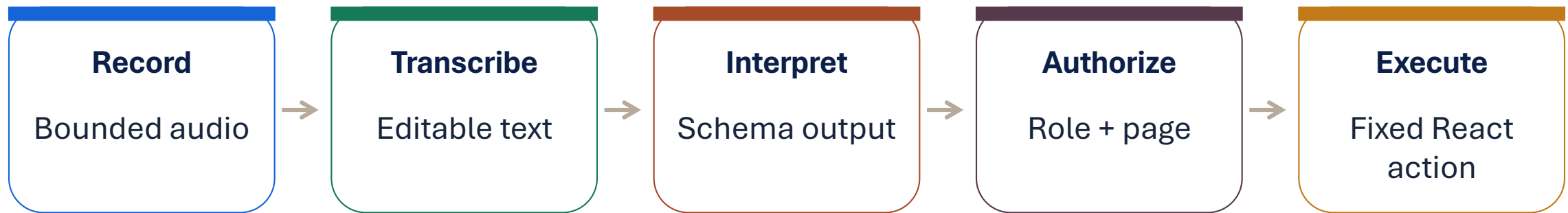
اسمي الأول مايا واسم العائلة حداد. رقم هاتفي 70123456 وأسكن في بيروت. اكتب في نبذة عني: أنا مؤثقة وصورة وأحب مساعدة الضيوف. أنهيت الثانوية. أستطيع القراءة والكتابة والعد باستقلالية. أريد عملاً مدفوعاً وبرنامج تدريب في الصفاة. أيّ الأنسب. أنا مهتمة بوظيفة كاشير وأعرف هذه الوظيفة

53/60 passed | 7 extraction defects retained | no automatic persistence

PROGRAMMATION

Voice navigation: flexible understanding, restricted execution

Natural language proposes an action; only the deterministic registry may authorize it.



Safety boundary

Unknown routes and cross-role actions are rejected. Submission and reset require confirmation.

Evidence and limits

57/57 registry checks passed. Noise and dictated email characters can still require correction.

Every voice operation keeps a keyboard and mouse equivalent.

Deterministic compatibility is explainable by construction

Controlled assessments produce reproducible task-level reasons.

Compatibility score

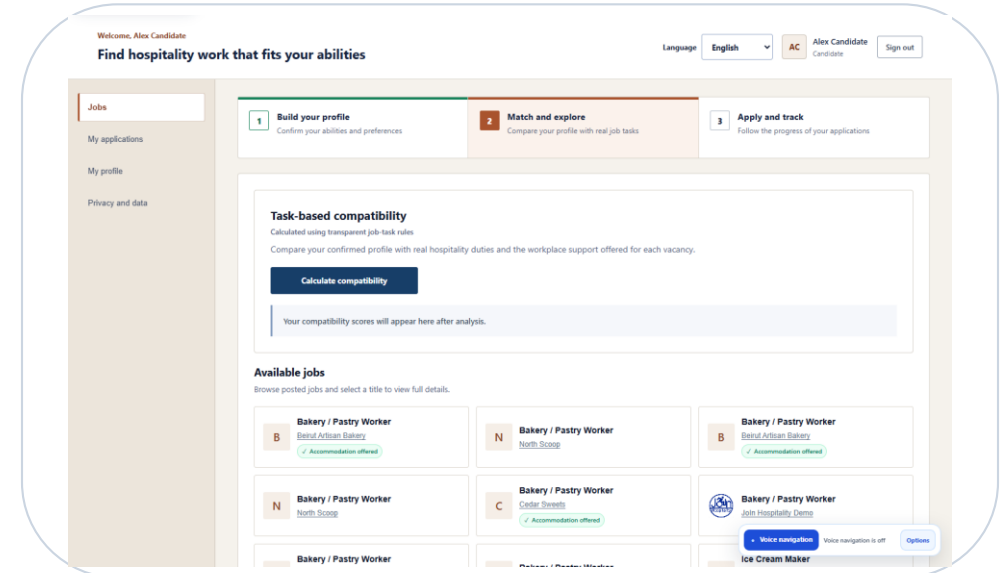
$$\frac{\text{Sum}(\text{adjusted weight} \times \text{feasibility factor})}{\text{Sum}(\text{adjusted weight}) \times 100}$$

Sum(adjusted weight) x 100

1.00 feasible

0.75 feasible with offered assistance

0.00 avoid; unsafe mandatory tasks remain gates



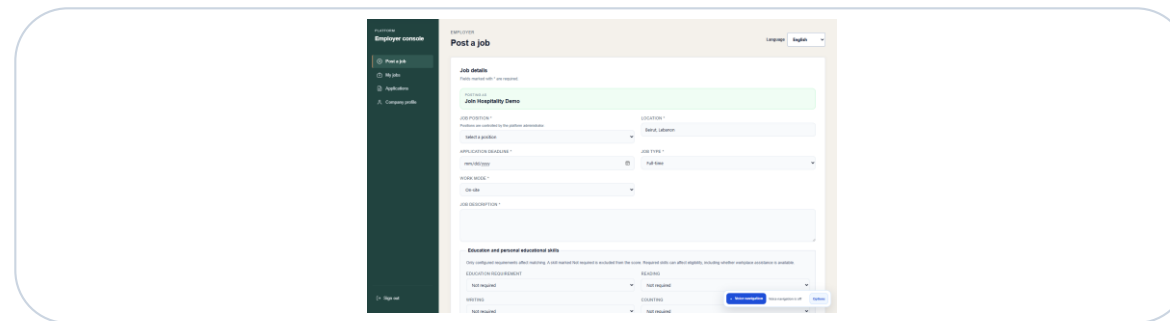
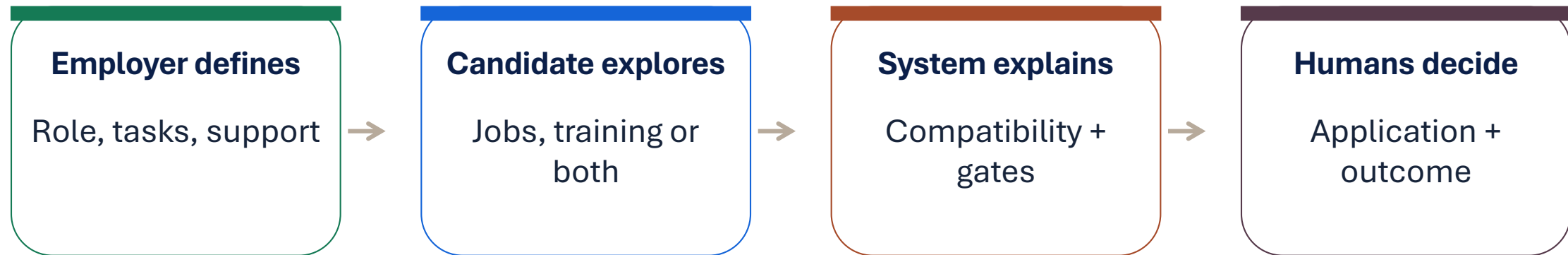
45 / 45

scoring cases passed

WORKFLOW INTEGRE

From controlled opportunity to auditable outcome

Candidates and employers contribute bounded choices; AI never creates a hiring decision.



Audit snapshot preserves score context and status history.

Iterative delivery from definition to evidence

A June-September plan controlled through versioned evidence and explicit risks.

JUN

Discover + define

Needs, stakeholders, FR / NFR

JUL

Design

Alternatives, UML / C4, prototype

JUL-AUG

Build + integrate

Catalogue, backend, AI, voice, scoring

AUG-SEP

Verify + deliver

Tests, users, staging, report, defence

Controls Focused Git commits | staging-deployment | render.yaml | requirement-to-test evidence | retained risks

Verification and validation across the lifecycle

Verification checks the specification; validation checks the user need.

Static + component

Build, lint, syntax, schema and unit checks

Service + integration

API workflows | scoring oracles | voice registry | AI matrix

System + staging

Browser E2E | security routes | performance | deployed origins

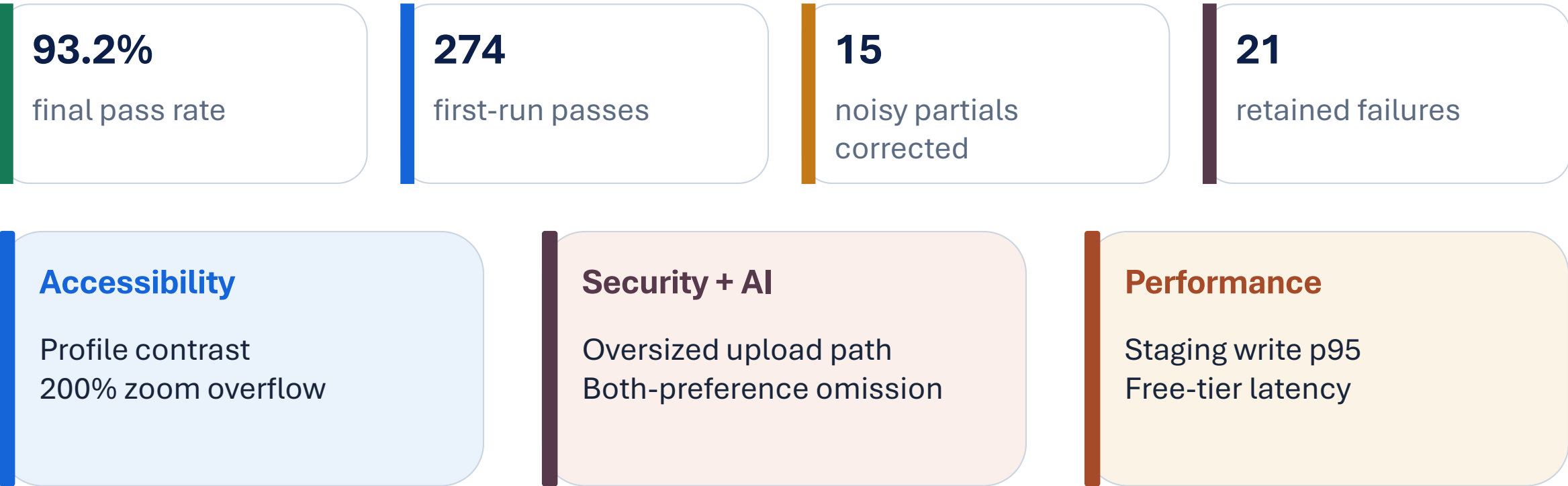
Human validation

Arc en Ciel session | JAWS workflow | stakeholder observation

Documentation: SRS -> architecture -> code/configuration -> test register -> evidence -> operational limits

310 cases completed; findings remained visible

The campaign records corrections and retained limitations, not only passes.

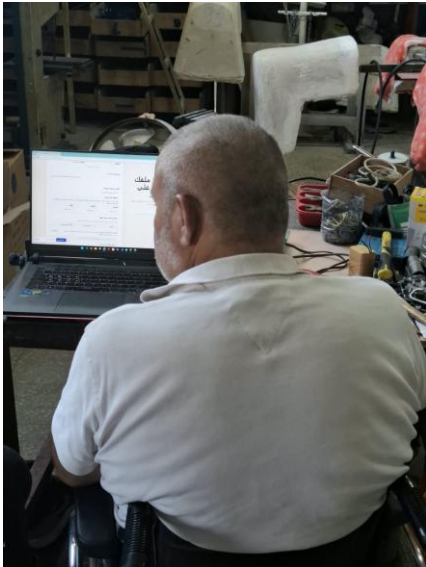


All 310 were executed; no case remained blocked or not run.

VALIDATION UTILISATEUR

The platform was evaluated with its intended users

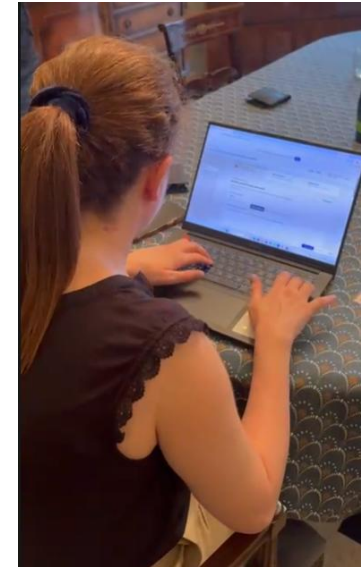
Human evidence supports tested users, not a universal accessibility claim.



Arabic workflow



20-person session



JAWS workflow

JAWS registration and application completed; open contrast and zoom findings remain reported.

Measured constraints guide the next engineering cycle

Latency and production governance define the remaining work.

Measured now

144/144 shared-load responses
100% functional availability
Latency target not consistent

Cost controls

Bottom-up by subsystem
AI and voice quotas
Measure before caching

Next cycle

Fix five material findings
Monitoring, backups and email
Repeat controlled user tests

Maintenance: corrective defects | adaptive change | perfective features | preventive refactoring

CONCLUSION

A validated bridge from ability to opportunity

Three ideas to retain.

1. Human control

AI suggests; candidates confirm; employers and verifiers decide.

2. Explainable engineering

Deterministic matching, explicit trust boundaries and auditable outcomes.

3. Evidence with limits

Cloud staging, 310 cases and human sessions - with open findings retained.

Questions?

Opportunity should be shaped by ability.